Vehicle Rental SaaS — Complete Booking Management Specification

Production-level specification for the complete booking lifecycle: availability, quotation, reservation, pricing, payment, deposit, verification, pickup, active rental, return, extension, cancellation, contracts, notifications and audit.

1. Booking Module — Main Sections

#	Section	Purpose
1	Booking Creation	Online, staff, phone, walk-in and partner bookings
2	Availability	Prevent conflicts and determine bookable inventory
3	Customer & Driver	Customer identity, eligibility and additional drivers
4	Vehicle Assignment	Category or specific vehicle reservation
5	Dates & Locations	Pickup/return schedule and branches
6	Pricing	Rates, discounts, extras, taxes and fees
7	Payments	Full/partial payments and payment status
8	Deposits	Security deposit collection, deductions and release
9	Documents	Required booking/customer documents
10	Contracts	Rental agreement and signatures
11	Pickup	Handover and initial inspection
12	Active Rental	Live rental management and extensions
13	Return	Final inspection and final charges
14	Cancellation	Cancellation, no-show and refunds
15	Notifications	Transactional and operational messages
16	Audit & Reports	History, analytics and operational reporting

2. Booking Creation Flow

- Select customer
- Select pickup and return locations
- Select pickup and return date/time
- Search availability
- Select vehicle category or specific vehicle
- Add extras
- Apply coupon/discount
- Calculate authoritative quote
- Verify customer/driver eligibility
- Verify required documents
- Select payment method
- Collect required payment/deposit
- Generate/sign contract if required
- Confirm booking and send notification

3. Booking Basic Information

- Booking ID and human-readable booking reference

- Booking source: online, staff, phone, walk-in, partner, corporate
- Booking status
- Customer ID and primary driver
- Assigned staff
- Created/updated timestamps
- Currency and timezone
- Internal notes and customer-facing notes

4. Dates & Times

- Pickup date/time
- Return date/time
- Pickup/return timezone
- Actual pickup and return timestamps
- Extension request time
- Calculated rental duration
- Grace period
- Late-return calculation

5. Locations

- Pickup branch/point
- Return branch/point
- One-way rental
- Delivery/collection address
- Delivery/collection fee
- Location business hours
- After-hours pickup/return
- Location-specific rules

6. Availability & Double-Booking Prevention

- Check confirmed bookings
- Check active rentals
- Check maintenance blocks
- Check damaged/unavailable vehicles
- Check vehicle/location constraints
- Check turnaround/buffer time
- Check booking status
- Recheck availability immediately before reservation
- Use backend transaction plus database locking/constraints
- Never rely on the React calendar alone

7. Vehicle Assignment

- Category reservation or specific vehicle reservation
- Requested category
- Assigned vehicle
- Assignment time and staff member

- Vehicle swap history
- Reason for reassignment
- Customer notification for meaningful changes

8. Centralized Pricing

- Base rental rate
- Vehicle/category override
- Rental duration
- Seasonal/weekend/holiday pricing
- Mileage policy and included kilometres
- Extra kilometre/hour charges
- Delivery/collection fees
- One-way fee
- Extras
- Coupon/discount
- Taxes
- Deposit
- Currency and rounding

9. Price Snapshot

- Store the complete authoritative pricing snapshot on the booking
- Base rental
- Adjustments
- Extras
- Discounts/coupons
- Tax
- Deposit
- Total
- Amount paid
- Amount due
- Refundable amount
- Historical bookings must not change when future pricing rules change

10. Booking Status Lifecycle

- Draft
- Quote
- Pending Verification
- Pending Payment
- Pending Confirmation
- Confirmed
- Pickup Scheduled
- Pickup In Progress
- Active Rental
- Return In Progress

- Completed
- Cancelled
- No Show
- Expired
- Rejected

11. Customer & Driver Eligibility

- Account status
- Email/phone verification
- KYC/identity status
- Driving licence validity
- Licence category
- Minimum age
- Minimum licence-holding period
- Additional driver requirements
- Blacklist/restriction status
- Required documents
- Deposit/payment requirement
- Repeat critical checks at pickup

12. Booking Documents

- Identity documents
- Driving licence
- Additional driver documents
- Signed forms
- Payment proof where needed
- Inspection/damage records
- Other booking-specific files
- Use central document management with versioning, verification, expiry and audit

13. Deposit Management

- Deposit amount
- Collection/authorization method
- Deposit status
- Receipt/reference
- Damage deductions
- Late/extra-charge deductions
- Partial deductions
- Refund amount
- Refund method
- Release date/reference
- Approval where required
- Idempotent processing

14. Payments

- Payment reference
- Payment method/provider
- Amount/currency
- Payment status
- Provider transaction ID
- Payment timestamp
- Failed-payment reason
- Partial payments
- Refund linkage
- Outstanding balance
- Design for eSewa, Khalti, Stripe and future providers

15. Extras / Add-ons

- Additional driver
- Child seat
- Helmet
- GPS
- Insurance add-on
- Delivery/collection
- Wi-Fi
- Extra-kilometre package
- Fuel package
- Custom business extras
- Each extra needs pricing, tax, quantity, availability and refund behavior

16. Pickup Workflow

- Confirm identity and licence
- Verify payment/deposit
- Confirm contract/signature
- Confirm assigned vehicle
- Record actual pickup time
- Record odometer
- Record fuel/battery
- Complete inspection
- Capture damage photos
- Record accessories
- Customer acknowledgment/signature
- Transition booking to active rental

17. Active Rental

- Live rental status
- Scheduled return
- Actual return when known
- Extension requests

- Additional payments
- Incidents/damage
- Overdue state
- Operational notes
- Automated reminders

18. Return Workflow

- Actual return date/time
- Odometer
- Fuel/battery
- Return inspection
- Pickup-vs-return condition comparison
- New damage
- Extra kilometres
- Extra hours/late fee
- Other charges
- Deposit deductions
- Final amount due
- Final payment
- Deposit release/refund
- Contract completion
- Vehicle availability update
- Maintenance scheduling
- Final receipt/notification

19. Extension / Modification

- New return date/time
- Availability recheck
- Recalculate price
- Additional payment/deposit if needed
- Contract update if required
- Approval workflow
- Notification
- Audit record

20. Cancellation & No-Show

- Who cancelled
- Reason
- Timestamp
- Policy applied
- Cancellation fee
- Refund amount
- Deposit handling
- Payment reversal/refund

- Inventory release
- Customer notification
- Audit history
- Keep no-show distinct when business rules differ

21. Reschedule / Modification

- Pickup date/time
- Return date/time
- Locations
- Vehicle category
- Assigned vehicle
- Drivers
- Extras
- Coupon/discount
- Notes
- Every change must revalidate permission, availability and price and preserve history

22. Contract Integration

- Contract template
- Booking data snapshot
- Customer and drivers
- Vehicle
- Pricing
- Deposit
- Terms
- Signature status
- Signed document
- Contract version
- Amendments

23. Notifications

- Booking created/confirmed
- Payment received/failed
- Documents missing
- Verification required
- Pickup reminder/instructions
- Return reminder
- Overdue return
- Extension approved/rejected
- Booking modified/cancelled
- Refund/deposit released
- Contract ready/signed
- Channels: email, SMS, WhatsApp, in-app

24. Booking Detail Page

- Overview
- Customer
- Driver(s)
- Vehicle
- Dates & locations
- Pricing
- Payments
- Deposit
- Documents
- Contract
- Pickup
- Active rental
- Return
- Damage/inspection
- Communication
- Notes
- Activity timeline
- Audit history

25. Booking Calendar & Search

- Day/week/month views
- Vehicle timeline
- Location timeline
- Category availability
- Booking status and pickup/return times
- Maintenance and blocked periods
- Search by reference, customer, phone/email, vehicle, driver, contract or payment reference
- Filter by status, date, location, category, vehicle, source, payment, verification, staff and overdue state
- Use server-side filtering/pagination

26. Operational Dashboard

- Today's bookings
- Today's pickups/returns
- Active rentals
- Upcoming bookings
- Pending confirmation
- Pending payments
- Pending verification
- Overdue returns
- Today's revenue
- Conversion/cancellation/no-show rates
- Fleet utilization
- Booking source distribution

27. Recommended Database Entities

- bookings
- booking_items
- booking_extras
- booking_status_history
- booking_modifications
- booking_vehicle_assignments
- booking_location_assignments
- booking_price_snapshots
- booking_cancellations
- booking_extensions
- booking_documents
- booking_notes
- booking_events
- availability_blocks
- payments
- payment_webhooks
- deposits
- refunds
- invoices
- contracts
- contract_signatures
- rental_checklists
- damage_records
- maintenance_work_orders
- notifications
- audit_logs

28. Booking Record — Minimum Fields

Field	Purpose
id	Unique booking ID
tenant_id	Tenant isolation
booking_reference	Human-friendly reference
customer_id / primary_driver_id	Customer and driver
source / status	Origin and lifecycle
pickup_location_id / return_location_id	Locations
pickup_at / return_at	Scheduled timestamps
actual_pickup_at / actual_return_at	Operational timestamps
vehicle_category_id / assigned_vehicle_id	Requested and assigned inventory
currency	Booking currency
subtotal / discount_total / tax_total	Pricing totals
deposit_amount / total_amount	Deposit and total

Field	Purpose
paid_amount / due_amount	Financial state
pricing_snapshot	Historical authoritative pricing
notes	Operational notes
created_by / assigned_staff_id	Ownership/assignment
created_at / updated_at	Audit timestamps
cancelled_at / completed_at	Lifecycle timestamps

29. State Machine Rules

- Draft → Quote → Pending Verification/Pending Payment → Confirmed → Pickup Scheduled → Pickup In Progress → Active Rental → Return In Progress → Completed.
- Confirmed may transition to Cancelled or No Show according to configured rules.
- Active Rental should not normally be cancelled except through an explicit administrative recovery workflow.
- Do not allow arbitrary status changes from the client.
- Every transition must be permission-checked, validated and audited.

30. API Areas

- POST /api/v1/bookings/quotes
- POST /api/v1/bookings
- GET /api/v1/bookings
- GET /api/v1/bookings/{id}
- PATCH /api/v1/bookings/{id}
- POST /api/v1/bookings/{id}/confirm
- POST /api/v1/bookings/{id}/cancel
- POST /api/v1/bookings/{id}/reschedule
- POST /api/v1/bookings/{id}/extend
- POST /api/v1/bookings/{id}/assign-vehicle
- POST /api/v1/bookings/{id}/pickup
- POST /api/v1/bookings/{id}/return
- GET /api/v1/availability
- GET /api/v1/bookings/{id}/price
- GET/POST /api/v1/bookings/{id}/payments
- GET/POST /api/v1/bookings/{id}/documents
- GET /api/v1/bookings/{id}/activity
- Public: /api/v1/public/{tenant}/availability, /pricing, /quotes, /bookings, /payments

31. Security & Multi-Tenant Rules

- Never trust tenant_id/company_id supplied by the browser.
- Every booking query and mutation must be tenant-scoped.
- Validate that customer, driver, vehicle, location and contract belong to the same tenant.
- Availability, pricing, payment state and rental state are backend authorities.
- Use idempotency for booking/payment operations that may be retried.
- Verify payment webhooks and provider signatures where supported.
- Protect private documents and internal notes.

- Do not expose sensitive payment, identity or risk information in logs.
- Audit important booking and financial changes.

32. Frontend Navigation

- Operations → Bookings
- Operations → Calendar
- Operations → Pickup & Return
- Operations → Active Rentals
- Operations → Availability
- Booking → New Booking
- Booking → Booking Detail
- Booking → Pending Verification
- Booking → Pending Payments
- Booking → Cancellations
- Booking → Extensions/Modifications
- Connected → Customers / Vehicles / Payments / Deposits / Refunds / Invoices / Contracts / Notifications / Reports

33. Recommended UI Rules

- Use a full page for New Booking and complex modifications.
- Use drawers for quick previews and simple edits.
- Use modals only for short confirmations/destructive actions.
- Keep a persistent price summary during booking creation/modification.
- Show availability before confirming a vehicle.
- Show state-specific primary actions.
- Use an activity timeline for operational/audit visibility.

34. Final Booking Checklist

■ Customer selected

■ Driver eligibility verified

■ Pickup/return locations valid

■ Dates/times valid

■ Backend availability rechecked

■ Vehicle/category assigned

■ Central pricing calculation completed

■ Pricing snapshot stored

■ Extras recorded

■ Required documents verified

■ Payment/deposit requirement satisfied

■ Contract generated/signed if required

■ Correct status transition performed

■ Confirmation notification sent

■ Audit record created

35. Most Important Architecture Rules

The booking domain is the core transaction of the rental platform. Backend/database systems must be authoritative for availability, pricing, tenant identity, payment state and rental state. Do not duplicate pricing logic between React and Go. Do not trust a vehicle ID merely because the browser selected it. Extensions, cancellations, pickup, return and rescheduling are complete business workflows—not simple status changes—and must validate inventory, money, documents, notifications and audit history.